Non-asymptotic Convergence guarantees for a Class of Optimization Problems

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Abstract

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I. Introduction

A. Notation

$\mathcal{M}(\Theta)$ and $\mathcal{M}_+(\Theta)$ denote the space of signed Borel measures and non-negative finite Borel measures on Θ, respectively. $\mathcal{P}(\Theta)$ is the space of Borel probability measures on Θ and $\mathcal{P}_2(\Theta)$ denotes its subset with finite second moment equipped with the 2-Wasserstein metric. δ_x denotes the Dirac measure at x. For a semiconvex function $f : \mathcal{X} \subseteq \mathbb{R}^m \to (-\infty, \infty]$, $\partial_x f$ denotes the standard convex subdifferential (or sub-gradient) of f with respect at x [Rockafellar(1970)]. For $\mu \in \mathcal{P}(\mathcal{X})$, $M_n(\mu) \coloneqq \int \|x\|_2^n \, d\mu(x)$ denotes its nth moment. $L^p(\mathcal{X}, \mu)$ denote the space of p-Lebesgue integrable functions supported on $\mathcal{X}$. For vector spaces V and W, $\mathcal{L}(\mathsf{V}, \mathsf{W})$ denotes the set of linear operators from V to W.

II. Problem Formulation

For $m, d \in \mathbb{N}$, let $\mathcal{X} \subseteq \mathbb{R}^m$, and $\Theta \subseteq \mathbb{R}^d$ denote the closure of a convex open set in $\mathbb{R}^d$. Let $\mathcal{G}(\Theta)$ be a separable Hilbert space of functions, formed via linear combinations from a large given parametrized set $\{\phi(\cdot, \theta) : \mathcal{X} \to \mathbb{R}\}_{\theta \in \Theta} \subseteq \mathcal{G}(\Theta)$. Denoting the corresponding Hilbert space norm by $\|\cdot\|$, we will assume that $\phi(\cdot, \theta)$ is strongly measurable and satisfies the growth condition

$$\|\phi(\cdot, \theta)\| \leq C_{\mathsf{Par}}\big(1 + \|\theta\|_2^2\big), \quad \forall\ \theta \in \Theta. \tag{1}$$

Let $\tilde{\mathcal{P}}_2(\Theta)$ denote a convex subset of $\mathcal{P}_2(\Theta)$. In the neural net optimization framework, Θ can be thought of as the space of parameters of a neural net, m is the dimension of the input to the neural net, and for $\rho \in \tilde{\mathcal{P}}_2(\Theta)$, $g_\rho(\cdot) \coloneqq \int_\Theta \phi(\cdot, \theta) d\rho(\theta)$ is the integral representation of the neural net output with respect to ρ. Note that the previous integral, interpreted in a Bochner sense, exists as an element of $\mathcal{G}(\Theta)$ since

$$\int_\Theta \|\phi(\cdot, \theta)\| \, d\rho(\theta) \leq \int C_{\mathsf{Par}}\big(1 + \|\theta\|_2^2\big) d\rho(\theta) \leq C_{\mathsf{Par}}\big(1 + M_2(\rho)\big) < \infty.$$

If the support of ρ is a finite discrete set, then the neural net output corresponds to that with a finite output width, and otherwise, to an infinite width neural net.

We are interested in the following regularized optimization problem with $\lambda, \tau \geq 0$:

$$J^\star\big(\tilde{\mathcal{P}}_2(\Theta), \lambda, \tau\big) \coloneqq \inf_{\rho \in \tilde{\mathcal{P}}_2(\Theta)} J(g_\rho, \lambda, \tau), \tag{2}$$

$$\text{where} \qquad J\big(\rho, \lambda, \tau\big) \coloneqq R(g_\rho) + \lambda \int V d\rho + \tau U(\rho), \tag{3}$$

where $R : \mathcal{G}(\Theta) \to \mathbb{R}_+$ denotes a cost function, and $V : \Theta \to \mathbb{R}^+$ and $U : \tilde{\mathcal{P}}_2(\Theta) \to \mathbb{R}$ are regularizers. Throughout, we will omit the dependence of V and U in the notation for the quantities defined henceforth, and also further simplify notation by dropping arguments when no confusion arises.

Remark 1 (Equivalence to a general problem) *The optimization problem above with $\tau = 0$ is equivalent to, up to a dimension and parameter space enlargement, to the more general problem in which $\tilde{\mathcal{P}}_2(\Theta)$ is replaced by $\mathcal{M}_+(\Theta)$ and more generally by $\mathcal{M}(\Theta)$, provided the regularizer V enables a suitable lifting to this enlarged space. This can be seen using similar arguments as in [Chizat and Bach(2018), Appendix A.2].*

We next state the regularity conditions assumed throughout the paper, which are similar to the ones in [Chizat and Bach(2018), Assumption 2.1].

Assumption 1 (Regularity conditions) *The following assumptions hold for Θ, $\tilde{\mathcal{P}}_2(\Theta)$, $\mathcal{G}(\Theta)$, ϕ, R, V and U:*

(i) *R is Fréchet differentiable at every $g_\rho \in \mathcal{G}(\Theta)$ with derivative $R'(g_\rho) \in \mathcal{L}\big(\mathcal{G}(\Theta), \mathbb{R}\big)$;*

(ii) *the Wasserstein gradient $\nabla_\theta \frac{\delta U}{\delta \rho}[\rho] \in L^2(\Theta, \rho)$ for every $\rho \in \tilde{\mathcal{P}}_2(\Theta)$;*

(iii) *$\phi(\cdot, \theta) : \Theta \to \mathcal{G}(\Theta)$ is (Fréchet) differentiable at every $\theta \in \Theta$ and the map $\theta \mapsto \phi(\cdot, \theta)$ is strongly measurable (with respect to $\|\cdot\|$) and satisfies (1);*

(iv) *V is proper, semiconvex, and lower semicontinuous, and U is geodesically semiconvex;*

(v) *there exists a family of $(Q_r)_{r>0}$ of nested non-empty bounded closed convex subsets of Θ such that*

(a) *$\{\theta \in \Theta : \mathsf{dist}(\theta, Q_r) \le r'\} \subseteq Q_{r+r'}$ for all $r, r' > 0$;*

(b) *for each Q_r, R' is uniformly Lipschitz on $\mathcal{G}_r(\Theta) := \{g_\rho : \rho \in \tilde{\mathcal{P}}_2(\Theta), \mathrm{supp}(\rho) \subseteq Q_r\}$ and uniformly bounded on sublevel sets of R contained in $\mathcal{G}_r(\Theta)$;*

(c) *on each Q_r, $\|\phi(\cdot, \theta)\|$, $V(\theta)$ are bounded and the Fréchet derivative $\phi'(\cdot, \theta) \in \mathcal{L}(\mathbb{R}^d, \mathcal{G}(\theta))$ is Lipschitz;*

(d) *there exists $c_1, c_2 > 0$ such that*

$$\sup_{\theta \in Q_r} \|\phi'(\cdot, \theta)\|_{\mathsf{op}} + \|\partial V(\theta)\|_{2,\mathsf{max}} + \left\| \nabla_\theta \frac{\delta U}{\delta \rho}[\rho](\theta) \right\|_2 \le c_1 + c_2 r,$$

for all $r > 0$ and $\rho \in \tilde{\mathcal{P}}_2(\Theta)$ such that $\mathrm{supp}(\rho) \subseteq Q_r$, where $\|\phi'(\cdot, \theta)\|_{\mathsf{op}}$ denotes the operator norm and $\|\partial V(\theta)\|_{2,\mathsf{max}}$ stands for the maximal Euclidean norm of an element in the set of subgradients $\partial V(\theta)$.

A. Random Finite Particle Objective

In practice, the optimization in (2) is performed via gradient descent over a subset of discrete probability measures, say with k atoms supported on Θ. For instance, this is the case when $\phi(\cdot, \theta)$ is a finite width neural net function with $\theta \in \Theta$ denoting the vector of all trainable parameters of the neural

net. In addition, R in the definition of J (see (3)) may not be known exactly, but would need to be estimated based on random observations Due to these discretizations and randomness of observations, the resulting estimate will only be an approximation of $J^\star\big(\tilde{\mathcal{P}}_2(\Theta), \lambda, \tau\big)$, in general. Formally, let $\hat{\mathcal{P}}_k(\Theta)$ denote the set of all discrete probability measures with at most k atoms supported on Θ, i.e., probability measures of the form $\left(\sum_{i=1}^n \delta_{\theta_i}\right)/k$.

Let $\mu \in \mathcal{P}(\mathcal{X})$ for some $\mathcal{X} \subseteq \mathbb{R}^m$, and X^n be i.i.d. observations distributed according to $\mu^{\otimes n}$. Set $\hat{\mu}_n \coloneqq \big(\sum_{i=1}^n \delta_{X_i}\big)/n$. Let $\mathcal{H}$ be a closed subspace of $L^2(\mathcal{X}, \mu)$ and $\bar{R} : \mathcal{H} \to L^2(\mathcal{X}, \mu)$. We will assume the following.

Assumption 2 (Regularity for Random Objective) *In place of Assumption 1 (i) and (v)-(b), we will assume that the following hold for Θ, $\tilde{\mathcal{P}}_2(\Theta)$, $\mathcal{G}(\Theta)$, and $\bar{R}$:*

(i) $\mathcal{G}(\Theta)$ is a closed subspace of the $\mathcal{H}$;

(ii) $\bar{R} : \mathcal{H} \to L^2(\mathcal{X}, \mu)$ is Fréchet differentiable at g_ρ for all $\rho \in \tilde{\mathcal{P}}_2(\Theta)$, with derivative $\bar{R}'(g_\rho) \in \mathcal{L}\big(\mathcal{H}, L^2(\mathcal{X}, \mu)\big)$.

(iii) for each Q_r, $\bar{R}'$ is uniformly Lipschitz on $\mathcal{G}_r(\Theta)$ and $g \mapsto \big\|\bar{R}'(g)\big\|_{\mathsf{op}}$ is uniformly bounded on sublevel sets of $g \mapsto \big\|\bar{R}(g)\big\|_{2,\mu}$ contained in $\mathcal{G}_r(\Theta)$.

Additionally, the remaining conditions in Assumption 1 pertaining to ϕ, V, U continues to hold.

Note that since $\bar{R} \circ g_\rho \in L^2(\mathcal{X}, \mu)$,

$$R_\mu(g_\rho) \coloneqq \mathbb{E}_\mu\big[\bar{R} \circ g_\rho\big] \quad \text{and} \quad \hat{R}_n(g_\rho) \coloneqq \mathbb{E}_{\hat{\mu}_n}\big[\bar{R} \circ g_\rho\big],$$

are well-defined. Assumption 2 (ii) then implies that R_μ is Fréchet differentiable at g_ρ for all $\rho \in \tilde{\mathcal{P}}_2(\Theta)$ with derivative $R'_\mu(g_\rho)[h] = \mathbb{E}_\mu\big[(\bar{R}'(g_\rho)[h]\big]$ for all $h \in \mathcal{H}$. Also, since $R'_\mu(g_\rho) \in L^2(\mathcal{X}, \mu)$ and $\big\|R'_\mu(g_\rho)\big\|_{2,\mu} \le \big\|\bar{R}'(g_\rho)\big\|_{\mathsf{op}}$, Assumption 2 (iii) implies Assumption 1 (v)-(b) with $R = R_\mu$. In other words, R_μ satisfies all the requirements of R in Assumption 1.

Considering the simpler case of $\tau = 0$, the relevant random finite particle objective is then

$$\hat{J}_n(\hat{\mathcal{P}}_k(\Theta), \lambda) \coloneqq \inf_{\rho \in \hat{\mathcal{P}}_k(\Theta)} \hat{J}_n(\rho, \lambda) \coloneqq \inf_{\rho \in \hat{\mathcal{P}}_k(\Theta)} \hat{R}_n(g_\rho) + \lambda \int V d\rho. \tag{4}$$

Note that $\mathbb{E}\big[\hat{J}_n(\rho, \lambda)\big] = R_\mu(g_\rho) + \lambda \int V d\rho$.

B. Particle and Wasserstein Gradient Flow

For $\boldsymbol{\theta} = (\theta_1, \ldots, \theta_k)$, define $\tilde{\rho}_k(\boldsymbol{\theta}) \coloneqq \big(\sum_{i=1}^k \delta_{\theta_i}\big)/k$, and let $\tilde{\rho}_{k,n,t} \coloneqq \tilde{\rho}_k\big(\boldsymbol{\theta}^{(n)}(t)\big) \in \hat{\mathcal{P}}_k(\Theta)$ denote the probability measure at time t of the gradient descent algorithm with the goal of minimizing the objective in (4). We will focus on the case where t takes continuous values in some interval, although discrete time formulations are also interesting in their own right. Defining $\hat{J}_{n,k}\big(\boldsymbol{\theta}^{(n)}(t)\big) \coloneqq$

$\hat{J}_n\big(\tilde{\rho}_{k,n,t}, \lambda\big)$, the evolution of $\tilde{\rho}_{k,n,t}$ under gradient descent can be viewed equivalently as the gradient flow of k particles, $\boldsymbol{\theta}^{(n)}(t) = \big(\theta_1^{(n)}(t), \ldots, \theta_k^{(n)}(t)\big)$ starting from some given initial state $\boldsymbol{\theta}^{(n)}(0)$ (see e.g. [Chizat and Bach(2018), Definition 2.2]).

Definition 1 (Particle gradient flow) *The k-particle gradient flow for the functional $\hat{J}_{n,k}$ is an absolutely continuous path $\boldsymbol{\theta} : \mathbb{R}_+ \to \Theta^k$ which satisfies $\theta_i'(t) \in -k\,\partial_{\theta_i}\hat{J}_{n,k}\big(\boldsymbol{\theta}(t)\big)$ for $i \in [1:k]$ and almost every $t \geq 0$. Note that gradient flows will select the subgradient with the smallest norm.*

Next, consider the gradient descent algorithm for finding the infimum in (2). Let $\rho_t \in \tilde{\mathcal{P}}_2(\Theta)$ denote the probability measures at time t of the descent algorithm. Analogous to particle gradient flow, the evolution of ρ_t can be viewed as a gradient flow of probability measures in the 2-Wasserstein space $(\mathcal{P}_2(\Theta), W_2)$.

Definition 2 (Wasserstein gradient flow) *A Wasserstein gradient flow for the functional J on a time interval $[0,T]$ is an absolutely continuous path $(\rho_t)_{t\in[0,T]}$ in $\mathcal{P}_2(\Theta)$ that satisfies the continuity equation distributionally on $[0,T]$, i.e.,*

$$\partial_t \rho_t = -\nabla \cdot \big(\rho_t v_t\big), \quad \textit{where} \quad v_t \in -\partial_\theta \left(\frac{\delta J}{\delta \rho}[\rho_t]\right), \tag{5}$$

where $v_t : \Theta \to \mathbb{R}^d$ denotes the velocity field.

C. Some Relevant Results

The deterministic setting where $\hat{J}_n\big(\rho, \lambda\big) = J\big(\rho, \lambda, 0\big)$ almost surely for all $n \in \mathbb{N}$ and $\rho \in \hat{\mathcal{P}}_k(\Theta)$ was studied in detail in [Chizat and Bach(2018)]. In this case, denoting $\boldsymbol{\theta}^{(n)}(t)$ by $\boldsymbol{\theta}(t)$ and $\hat{J}_{n,k}\big(\boldsymbol{\theta}^{(n)}(t), \lambda\big)$ by $J_k\big(\boldsymbol{\theta}(t), \lambda\big)$, the following theorem summarizing (in a simplified form) Proposition 2.3, Proposition 2.5, Theorem 2.6 in [Chizat and Bach(2018)] states the uniqueness of particle and Wasserstein gradient flows defined above. In addition, the authors also show that the particle gradient flows converge to a corresponding Wasserstein gradient flow (in 2-Wasserstein metric for all t) in the mean field limit $k \to \infty$.

Theorem 1 (Proposition 2.3, Proposition 2.5 and Theorem 2.6 in [Chizat and Bach(2018)]) *Let $\tau = 0$ and $\tilde{\mathcal{P}}_{2,\mathsf{Leb}}(\Theta)$ denote the set of all Lebesgue (absolutely continuous) densities on Θ. For any initial state $\boldsymbol{\theta}(0) \in \Theta^k$, there is a unique particle gradient flow for the functional J_k. Similarly, under Assumption 1, for any initialization $\rho_0 \in \tilde{\mathcal{P}}_{2,\mathsf{Leb}}(\Theta)$ supported on $Q_{r_0} \subseteq \Theta$, there exists a unique Wasserstein gradient flow for J. Let $(t \mapsto \boldsymbol{\theta}_k(t))_{k\in\mathbb{N}}$ be a sequence of particle gradient flows for $(J_k)_{k\in\mathbb{N}}$ initialized in a set Q_{r_0}. If $\tilde{\rho}_k\big(\boldsymbol{\theta}(0)\big)$ converges to ρ_0 in W_2, then $\tilde{\rho}_k\big(\boldsymbol{\theta}(t)\big)$ converges as $k \to \infty$, to the unique Wasserstein gradient flow of J starting from ρ_0.*

It was also shown in [Chizat and Bach(2018)] that under certain additional regularity assumptions on ϕ, V, R and Θ (which includes convexity of R), if the gradient flow in the mean field limit converges, then it does converge to the global minimizer of J.

Theorem 2 (Theorem 3.3 and Theorem 3.5 in [Chizat and Bach(2018)]) *Suppose the regularity assumptions on ϕ, V, R and Θ and the initialization ρ_0 stated in [Chizat and Bach(2018), Theorem 3.3 and 3.5] hold. If ρ_t converges to some ρ_∞ in W_2, then ρ_∞ is a global minimizer of J.*

D. Regularized Wasserstein Gradient Flow and Langevin Particle Descent

Consider $\tau > 0$ and let $\tilde{\mathcal{P}}_2(\Theta)$ be a subset of $\tilde{\mathcal{P}}_{2,\mathsf{Leb}}(\Theta)$ such that $U(\rho)$ is well defined for all $\rho \in \tilde{\mathcal{P}}_2(\Theta)$. It can be shown that there exists a unique Wasserstein gradient flow $(\rho_t)_t$, $t \in \mathbb{R}_+$, for J satisfying the continuity equation (5). This follows using similar arguments as in [Chizat and Bach(2018), Proof of Proposition 2.5] under the additional assumptions on U such as $\left\|\nabla_\theta \frac{\delta U}{\delta \rho}[\rho](\theta)\right\|_2 \leq c_1 + c_2 r$ on Q_r given in Assumption 1. Specifically, it holds that for any T and $t \in [0, T)$, $\mathrm{supp}(\rho_t)$ is contained in a bounded subset $Q_{r(T)}$ for some $r(T) < \infty$. Moreover, Theorem 1 can also be proven to hold.

There are several standard choices for U. Among these, we will be concerned mostly with $U(\rho) = H(\rho)$, where $H(\rho)$ is the (negative differential) entropy defined as

$$H(\rho) := \begin{cases} \int_\Theta \rho \log \rho \, d\theta, & \text{if} \quad \rho \ll \mathsf{Leb}, \\ \infty, & \text{otherwise}, \end{cases} \tag{6}$$

for ρ such that integral is defined. By abuse of notation, ρ denotes both the measure and its Lebesgue density in the above definition, and we will follow this henceforth. In this case, we have

$$\frac{\delta H}{\delta \rho}[\rho] = 1 + \log \rho, \qquad \nabla_\theta \frac{\delta H}{\delta \rho}[\rho] = \frac{\nabla_\theta \rho}{\rho}. \tag{7}$$

Note that

$$\left\|\nabla_\theta \frac{\delta H}{\delta \rho}[\rho]\right\|_{2,\rho}^2 = \int_\Theta \frac{\|\nabla \rho\|_2^2}{\rho} =: I(\rho), \tag{8}$$

where $I(\rho)$ is the Fisher information at ρ.

The relevant Wasserstein gradient flow in the mean field limit is

$$\partial_t \rho_t = -\nabla \cdot \big(\rho_t v_t\big), \quad \text{where} \quad v_t(\theta) \in -\partial_\theta \left(\frac{\delta J}{\delta \rho}[\rho_t](\theta)\right), \tag{9}$$

$$\text{i.e.} \qquad v_t(\theta) = -\left(\mathbb{E}_\mu\big[\bar{R}'(g_{\rho_t}) \nabla_\theta \phi(\cdot, \theta)\big] + \tau \frac{\nabla_\theta \rho}{\rho}\right) - \xi(\theta), \quad \xi(\theta) \in \partial V(\theta).$$

Along the Wasserstein gradient flow ρ_t with $\mathrm{supp}(\rho_t) \subseteq Q_r$ for some r, Assumption 1 implies that

$$I(\rho_t) \coloneqq \int_\Theta \frac{\|\nabla \rho_t\|_2^2}{\rho_t} d\theta = \int_\Theta \left\| \nabla_\theta \frac{\delta H}{\delta \rho}[\rho_t] \right\|_2^2 d\rho_t \leq (c_1 + c_2 r)^2.$$

The evolution of ρ_t in (9) corresponds to the law of $\theta(t)$ that satisfy the stochastic differential equation (Langevin diffusion):

$$d\theta(t) \coloneqq -\mathbb{E}_\mu \big[R'(g_{\rho_t}) \nabla_\theta \phi\big(\cdot, \theta(t)\big)\big] dt - \xi\big(\theta(t)\big) dt + \sqrt{2\tau}\, dB(t), \quad \xi\big(\theta(t)\big) \in \partial V\big(\theta(t)\big), \tag{10}$$

where $B(t)$ is the standard d-dimensional Wiener process (or Brownian motion).

Let $\tilde{\rho}_{k,n,t}$ denote the empirical probability measure of $\boldsymbol{\theta}^{(n)}(t)$ at time t corresponding to particle gradient flow starting from $\boldsymbol{\theta}^{(n)}(0) = \bar{\boldsymbol{\theta}}(0)$. The relevant random finite particle evolution is governed by the Langevin diffusion:

$$\begin{aligned} d\theta_i^{(n)}(t) = &- \mathbb{E}_{\hat{\mu}_n} \big[\bar{R}'(g_{\tilde{\rho}_{k,n,t}}) \nabla_\theta \phi\big(\cdot, \theta_i^{(n)}(t)\big)\big] dt - \xi\big(\theta_i^{(n)}(t)\big) dt + \sqrt{2\tau}\, dB_i(t), \\ &\xi\big(\theta_i^{(n)}(t)\big) \in \partial V\big(\theta_i^{(n)}(t)\big), \end{aligned} \tag{11}$$

where $\{B_i(t)\}_{i=1}^k$ are k independent copies of $B(t)$.

E. Key Questions

Given the above background, some pertinent questions remain as follows:

(i) Can the optimization error, defined as the difference of the (non-regularized) empirical and mean field objectives along the particle and wasserstein gradient flows at time t:

$$\left| \hat{R}_{n,k,t} - R_{\mu,t} \right| \tag{12}$$

where $\hat{R}_{n,k,t} \coloneqq \hat{R}_n\big(\tilde{\rho}_{k,n,t}\big)$ and $R_{\mu,t} \coloneqq R_\mu(\rho_t)$, be quantified in terms of k, n, t, λ and τ ?

(ii) What are the conditions under which ρ_t converges to some ρ_∞ in W_2 and what is the rate of convergence ?

III. Main results

A. Optimization Error Bounds

Let

$$\tilde{\mathcal{G}} \coloneqq \{g_\rho : \rho \in \tilde{P}_2(\Theta)\}, \tag{13a}$$

$$\bar{R} \circ \tilde{\mathcal{G}} \coloneqq \big\{\bar{R}(g_\rho) : \rho \in \tilde{\mathcal{P}}_2(\Theta)\big\} \subseteq L^2(\mathcal{X}, \mu). \tag{13b}$$

Suppose there exists a measurable envelope[1] $\tilde{G}$ for $\tilde{\mathcal{G}}$ and $\tilde{G}_{\bar{R}}$ for $\bar{R}\circ\tilde{\mathcal{G}}$. For a function class $\mathcal{F}$ with measurable envelope F, let

$$E\big(\mathcal{F},F\big) \coloneqq \int_0^1 \sup_Q \sqrt{1+\log N\big(\mathcal{F},\|\cdot\|_{2,Q},\epsilon\big\|F\big\|_{2,Q}\big)}d\epsilon, \tag{14}$$

be the (Dudley's) entropy integral of $\mathcal{F}$ with respect to envelope F, where $N\big(\mathcal{F},\|\cdot\|_{2,\gamma},\epsilon\big)$ is the ϵ-covering number of $\mathcal{F}$ in $\|\cdot\|_{2,\gamma}$ for $\gamma\in\mathcal{P}(\mathcal{X})$, and the supremum above is over all finitely supported discrete probability measures Q on $\mathcal{X}$.

We have the following result which provides an upper bound on the expected optimization error given in (12).

Theorem 3 (Optimization Error Bounds) *Let $\mu\in\mathcal{P}_2(\mathcal{X})$ and $\tilde{\mathcal{P}}_2(\Theta)=\tilde{\mathcal{P}}_{2,\mathsf{Leb},H}(\Theta,C_{\mathsf{M}})$ denote the subset of $\mathcal{P}_2(\Theta)$ with Lebesgue densities, finite entropy $H(\rho)$, finite Fisher information $I(\rho)$, and $M_2(\rho)\le C_{\mathsf{M}}$ for $C_{\mathsf{M}}>0$. Suppose the following hold:*

(i) Assumption 2 is satisfied;

(ii) $\phi(x,\cdot):\Theta\to\mathbb{R}$ as well as $\nabla_\theta\phi(x,\cdot):\Theta\to\mathbb{R}^d$ are Lipschitz with Lipschitz constant bounded by $C_{\mathsf{L}}\|x\|$. Moreover, $\partial_j\phi(\cdot,\theta)\in\mathcal{H}\subseteq L^2(\mathcal{X},\mu)$ for all $\theta\in\Theta$ and $j\in[1:d]$;

(iii) $\bar{R}\circ\tilde{\mathcal{G}}$ has a measurable envelope $\tilde{G}_{\bar{R}}$ such that $\tilde{G}_{\bar{R}}\in L^2(\mathcal{X},\mu)$ and $E\big(\bar{R}\circ\tilde{\mathcal{G}},\tilde{G}_{\bar{R}}\big)<\infty$. There exists $L_{\bar{R},\tilde{\mathcal{G}}}\in L^2(\mathcal{X},\mu)$ such that for μ a.e.,

$$\big|\bar{R}(g)-\bar{R}(\tilde{g})\big|\le L_{\bar{R},\tilde{\mathcal{G}}}\,|g-\tilde{g}|\,,\ \forall g,\tilde{g}\in\tilde{\mathcal{G}}.$$

Also, there exists $L_{\bar{R}',\tilde{\mathcal{G}}},G_{\bar{R}',\tilde{\mathcal{G}}}\in L^2(\mathcal{X},\mu)$ such that $\bar{R}'(g)\in\mathcal{L}\big(\mathcal{H},L^2(\mathcal{X},\mu)\big)$ satisfies

$$\begin{aligned}
&|R'(g)[h]|\le G_{R',\tilde{\mathcal{G}}}\,|h|\,,\quad \forall\ g\in\tilde{\mathcal{G}}\quad \textit{and}\quad h\in\mathcal{H},\\
&\big|\bar{R}'(g)[h]-\bar{R}'(\tilde{g})[h]\big|\le L_{\bar{R}',\tilde{\mathcal{G}}}\,|g-\tilde{g}|\,|h|\,,\quad \forall\ g,\tilde{g}\in\tilde{\mathcal{G}}\quad \textit{and}\quad h\in\mathcal{H}.
\end{aligned}\tag{15}$$

Then

$$\begin{aligned}
\mathbb{E}\left[\left|\hat{R}_{n,k,t}-R_{\mu,t}\right|\right]\le\big\|L_{\bar{R},\tilde{\mathcal{G}}}\big\|_{2,\mu}\sqrt{M_2(\mu)}\left(\sqrt{\frac{C_{\mathsf{M}}}{k}}+C_{\mathsf{L}}\left(\frac{\sqrt{\bar{c}_{\mu,\tilde{G}}}}{\sqrt{n}}+\frac{\sqrt{\tilde{c}_{\mu,\tilde{G}}}}{\sqrt{k}}\right)e^{t\,c_{\mu,\tilde{G}}}\right)\\
+\frac{C_{\mathsf{D}}}{\sqrt{n}}E\big(\bar{R}\circ\tilde{\mathcal{G}},\tilde{G}_{\bar{R}}\big)\big\|\tilde{G}_{\bar{R}}\big\|_{2,\mu},
\end{aligned}\tag{16}$$

where

$$c_{\mu,\tilde{G}}\coloneqq 0.5\Big(1+2\beta_V+6C_{\mathsf{L}}^2\,M_2(\mu)\big\|G_{\bar{R}',\tilde{\mathcal{G}}}\big\|_{2,\mu}^2+6C_{\mathsf{L}}^4\left\|L_{\bar{R}',\tilde{\mathcal{G}}}\,\|X\|_2\right\|_{2,\mu}^2M_2(\mu)\Big),\tag{17a}$$

[1]An envelope of a class $\mathcal{F}$ of functions $f:\mathcal{X}\to\mathbb{R}$ is a function $F:\mathcal{X}\to[0,\infty]$ such that $|f(x)|\le F(x)$ for all $f\in\mathcal{F}$ and $x\in\mathcal{X}$.

$$\tilde{c}_{\mu,\tilde{G}} \coloneqq \frac{12\, C_{\mathsf{L}}^4 C_{\mathsf{M}} \big\| L_{\bar{R}',\tilde{\mathcal{G}}} \big\|_{2,\mu}^2 M_4(\nu)}{c_{\mu,\tilde{G}}}, \qquad \bar{c}_{\mu,\tilde{G}} \coloneqq \frac{3C_{\mathsf{L}}^2 \mathbb{E}_\mu \Big[G_{\bar{R}',\tilde{\mathcal{G}}}^2 \, \|X\|_2^2 \Big]}{2c_{\mu,\tilde{G}}}, \tag{17b}$$

and β_V is such that V is β-semiconvex.

Proof. We have

$$\begin{aligned} \Big| \hat{R}_{n,k,t} - R_{\mu,t} \Big| &\coloneqq \Big| \hat{R}_n\big(g_{\tilde{\rho}_{k,n,t}}\big) - R_\mu(g_{\rho_t}) \Big| \\ &\le \Big| \hat{R}_n\big(g_{\tilde{\rho}_{k,n,t}}\big) - R_\mu(g_{\tilde{\rho}_{k,n,t}}) \Big| + \big| R_\mu(g_{\rho_t}) - R_\mu(g_{\tilde{\rho}_{k,n,t}}) \big| \,. \end{aligned} \tag{18}$$

Note that $g_{\tilde{\rho}_{k,n,t}}$ depends on $\hat{\mu}_n$, in general, due to which upper bounding the expectation of the second term is not straightforward. To avoid this issue, taking supremum over $f \in \bar{R} \circ \tilde{\mathcal{G}}$,

$$\begin{aligned} \Big| \hat{R}_n\big(g_{\tilde{\rho}_{k,n,t}}\big) - R_\mu(g_{\tilde{\rho}_{k,n,t}}) \Big| &= \big| \mathbb{E}_{\hat{\mu}_n}[\bar{R} \circ g_{\tilde{\rho}_{k,n,t}}] - \mathbb{E}_\mu[\bar{R} \circ g_{\tilde{\rho}_{k,n,t}}] \big| \\ &\le \sup_{f \in \bar{R} \circ \tilde{\mathcal{G}}} |\mathbb{E}_{\hat{\mu}_n}[f] - \mathbb{E}_\mu[f]| =: \|\hat{\mu}_n - \mu\|_{\bar{R} \circ \tilde{\mathcal{G}}} \,, \end{aligned}$$

where $\|\hat{\mu}_n - \mu\|_{\mathcal{F}} := \sup_{f \in \mathcal{F}} |\mathbb{E}_{\hat{\mu}_n}[f] - \mathbb{E}_\mu[f]|$. Taking expectations, we obtain via Dudley's entropy integral bound that

$$\mathbb{E}\left[\big| \mathbb{E}_{\hat{\mu}_n}[\bar{R} \circ g_{\tilde{\rho}_{k,n,t}}] - \mathbb{E}_\mu[\bar{R} \circ g_{\tilde{\rho}_{k,n,t}}] \big| \right] \le \mathbb{E}\big[\, \|\hat{\mu}_n - \mu\|_{\bar{R} \circ \tilde{\mathcal{G}}} \, \big] \le \frac{C_{\mathsf{D}} \, E\big(\bar{R} \circ \tilde{\mathcal{G}}, \tilde{G}_{\bar{R}}\big) \big\| \tilde{G}_{\bar{R}} \big\|_{2,\mu}}{\sqrt{n}}, \tag{19}$$

where $E\big(\bar{R} \circ \tilde{\mathcal{G}}, \tilde{G}_{\bar{R}}\big)$ is as defined in (14) and C_{D} is a universal constant.

For the second term in (18), an application of Cauchy-Schwarz inequality yields

$$\big| \mathbb{E}_\mu[\bar{R} \circ g_{\rho_t}] - \mathbb{E}_\mu[\bar{R} \circ g_{\tilde{\rho}_{k,n,t}}] \big| \le \big\| L_{\bar{R},\tilde{\mathcal{G}}} \, \big| g_{\rho_t} - g_{\tilde{\rho}_{k,n,t}} \big| \, \big\|_{1,\mu} \le \big\| L_{\bar{R},\tilde{\mathcal{G}}} \big\|_{2,\mu} \big\| g_{\rho_t} - g_{\tilde{\rho}_{k,n,t}} \big\|_{2,\mu} .$$

Hence

$$\begin{aligned} &\mathbb{E}\left[\big| \mathbb{E}_{\hat{\mu}_n}[\bar{R} \circ g_{\tilde{\rho}_{k,n,t}}] - \mathbb{E}_\mu[\bar{R} \circ g_{\tilde{\rho}_{k,n,t}}] \big| \right] \\ &\le \frac{C_{\mathsf{D}} \, E\big(\bar{R} \circ \tilde{\mathcal{G}}, \tilde{G}_{\bar{R}}\big) \big\| \tilde{G}_{\bar{R}} \big\|_{2,\mu}}{\sqrt{n}} + \big\| L_{\bar{R},\tilde{\mathcal{G}}} \big\|_{2,\mu} \mathbb{E}\Big[\, \big\| g_{\rho_t} - g_{\tilde{\rho}_{k,n,t}} \big\|_{2,\mu} \Big]. \end{aligned} \tag{20}$$

For bounding the last term in the equation above, we will use a propagation of chaos inspired technique. To this end, let $\bar{\boldsymbol{\theta}}(t) = \big(\bar{\theta}_1(t), \ldots, \bar{\theta}_k(t)\big) \sim \rho_t^{\otimes k}$ such that the particles $\{\bar{\theta}_i(t)\}_{i=1}^k$ each evolve independently at time t starting from $\bar{\boldsymbol{\theta}}(0) \sim \rho_0^{\otimes k}$ according to the mean field dynamics (10):

$$d\theta_i(t) \coloneqq -\mathbb{E}_\mu\big[R'(g_{\rho_t}) \nabla_\theta \phi\big(\cdot, \theta_i(t)\big) \big] dt - \xi\big(\theta_i(t)\big) dt + \sqrt{2\tau} \, dB_i(t), \quad \xi(\theta_i(t)) \in \partial V\big(\theta_i(t)\big), \tag{21}$$

where $\{B_i(t)\}_{i=1}^k$ is the same coupled Brownian motion as in (11).

Let $\bar{\rho}_{k,t}(\bar{\boldsymbol{\theta}}(t)) = \big(\sum_{i=1}^k \delta_{\bar{\theta}_i(t)} \big)/k$ denote the empirical distribution of $\bar{\boldsymbol{\theta}}(t)$. Using triangle inequality,

$$\big\| g_{\rho_t} - g_{\tilde{\rho}_{k,n,t}} \big\|_{2,\mu} \le \big\| g_{\rho_t} - g_{\bar{\rho}_{k,t}} \big\|_{2,\mu} + \big\| g_{\bar{\rho}_{k,t}} - g_{\tilde{\rho}_{k,n,t}} \big\|_{2,\mu} . \tag{22}$$

Since $\{\bar{\theta}_i\}_{i=1}^k$ are i.i.d.,

$$\begin{aligned}
\mathbb{E}_{\rho_t^{\otimes k}}\left[\left\|g_{\rho_t}-g_{\bar{\rho}_{k,t}}\right\|_{2,\mu}^2\right] &= \int_{\mathcal{X}} \mathbb{E}_{\rho_t^{\otimes k}}\left[\left(g_{\bar{\rho}_{k,t}}(x)-g_{\rho_t}(x)\right)^2\right] d\mu(y) \\
&= \frac{1}{k^2}\sum_{i=1}^k \mathbb{E}_{\mu\otimes\rho_t}\left[\left(\phi(X,\bar{\theta}_i)-\mathbb{E}_{\rho_t}\left[\phi(X,\bar{\theta})\right]\right)^2\right] \\
&\overset{(a)}{=} \frac{1}{k}\mathbb{E}_{\mu\otimes\rho_t}\left[\left(\phi(X,\bar{\theta}_i)-\mathbb{E}_{\rho_t}\left[\phi(X,\bar{\theta})\right]\right)^2\right] \\
&\leq \frac{1}{k}\mathbb{E}_{\mu\otimes\rho_t^{\otimes 2}}\left[\left(\phi(X,\bar{\theta}_i)-\phi\big(X,\bar{\theta}\,\big)\right)^2\right] \\
&\leq \frac{1}{k}\mathbb{E}_{\mu\otimes\rho_t^{\otimes 2}}\left[\|X\|_2^2\left\|\bar{\theta}_i-\bar{\theta}\right\|_2^2\right] \\
&= \frac{1}{k}\mathbb{E}_{\mu}\left[\,\|X\|_2^2\,\right]\mathbb{E}_{\rho_t^{\otimes 2}}\left[\,\left\|\bar{\theta}_i-\bar{\theta}\right\|_2^2\,\right] \\
&\leq \frac{C_{\mathsf{M}}M_2(\mu)}{k},
\end{aligned} \tag{23}$$

where in (a), we used for $i\neq j$,

$$\begin{aligned}
&\int_{\mathcal{X}} \mathbb{E}_{\rho_t^{\otimes 2}}\left[\left(\phi(x,\bar{\theta}_i)-\mathbb{E}_{\rho_t}\left[\phi(x,\bar{\theta})\right]\right)\left(\phi(x,\bar{\theta}_j)-\mathbb{E}_{\rho_t}\left[\phi(x,\bar{\theta})\right]\right)\right] d\mu(x) \\
&= \int_{\mathcal{X}} \mathbb{E}_{\rho_t}\left[\phi(x,\bar{\theta}_i)-\mathbb{E}_{\rho_t}\left[\phi(x,\bar{\theta})\right]\right]\mathbb{E}_{\rho_t}\left[\phi(x,\bar{\theta}_j)-\mathbb{E}_{\rho_t}\left[\phi(x,\bar{\theta})\right]\right] d\mu(x) = 0.
\end{aligned} \tag{24}$$

Considering the second term in the RHS of (22) and using Lipschitzness of $\phi(x,\cdot)$,

$$\begin{aligned}
\left\|g_{\bar{\rho}_{k,t}}-g_{\tilde{\rho}_{k,n,t}}\right\|_{2,\mu}^2 &= \int_{\mathcal{X}}\left(g_{\bar{\rho}_{k,t}}(x)-g_{\tilde{\rho}_{k,n,t}}(x)\right)^2 d\mu(x) \\
&= \int_{\mathcal{X}}\left(\frac{1}{k}\sum_{i=1}^k\left[\phi\big(x,\bar{\theta}_i(t)\big)-\phi\big(x,\bar{\theta}_i^{(n)}(t)\big)\right]\right)^2 d\mu(y) \\
&\leq \frac{C_{\mathsf{L}}^2}{k}\int_{\mathcal{X}}\|x\|^2 d\mu(y)\left(\sum_{i=1}^k\left\|\bar{\theta}_i(t)-\bar{\theta}_i^{(n)}(t)\right\|^2\right) \\
&\leq C_{\mathsf{L}}^2 M_2(\mu)e_k^2(t),
\end{aligned}$$

where

$$e_k^2(t) \coloneqq \frac{1}{k}\left(\sum_{i=1}^k\left\|\bar{\theta}_i(t)-\bar{\theta}_i^{(n)}(t)\right\|^2\right). \tag{25}$$

By Jensen's inequality,

$$\mathbb{E}\left[\left\|g_{\rho_t}-g_{\tilde{\rho}_{k,n,t}}\right\|_{2,\mu}\right] \leq \sqrt{\mathbb{E}\left[\left\|g_{\rho_t}-g_{\tilde{\rho}_{k,n,t}}\right\|_{2,\mu}^2\right]} \leq C_{\mathsf{L}}\sqrt{M_2(\mu)\mathbb{E}\big[e_k^2(t)\big]}. \tag{26}$$

The next lemma obtains a bound on $\mathbb{E}\big[e_k^2(t)\big]$.

Lemma 4 *[Propagation of Chaos inspired Error Bound] Under the assumptions in Theorem 3,*

$$\mathbb{E}\big[e_k(t)^2\big] \leq \left(\frac{\tilde{c}_{\mu,\tilde{G}}}{k}+\frac{\bar{c}_{\mu,\tilde{G}}}{n}\right)e^{2t\,c_{\mu,\tilde{G}}}, \tag{27}$$

where $c_{\mu,\tilde{G}}$, $\tilde{c}_{\mu,\tilde{G}}$, and $\bar{c}_{\mu,\tilde{G}}$ are as defined in (17).

From (20), (26) and (27), the claim in (16) follows using $\sqrt{x+y} \le \sqrt{x} + \sqrt{y}$ for all $x, y \ge 0$. □

B. *Proof of Lemma 4*

Let

$$b_j(\theta,\rho) \coloneqq -\mathbb{E}_\mu\big[\bar{R}'(g_\rho)\partial_j\phi(\cdot,\theta)\big],$$
$$\hat{b}_{n,j}(\theta,\rho) \coloneqq -\mathbb{E}_{\hat{\mu}_n}\big[\bar{R}'(g_\rho)\partial_j\phi(\cdot,\theta)\big].$$

We have from (11) and (21) that

$$d\bar{\theta}_{i,j}(t) - d\theta^{(n)}_{i,j}(t) = \Big(b_j(\bar{\theta}_i(t),\rho_t) - \hat{b}_{n,j}\big(\theta^{(n)}_i(t),\tilde{\rho}_{k,n,t}\big)\Big)dt - \Big(\xi_j\big(\bar{\theta}_i(t)\big) - \xi_j\big(\theta^{(n)}_i(t)\big)\Big).$$

Hence, taking derivatives of $\mathbb{E}\big[e_k(t)^2\big]$ with respect to t, we obtain

$$\begin{aligned}
\frac{d}{dt}\mathbb{E}\big[e_k(t)^2\big] &= \frac{2}{k}\sum_{i=1}^{k}\sum_{j=1}^{d}\Big[\mathbb{E}\Big[\big(\bar{\theta}_{i,j}(t) - \bar{\theta}^{(n)}_{i,j}(t)\big)\big(b_j(\bar{\theta}_i(t),\rho_t) - \hat{b}_{n,j}\big(\theta^{(n)}_i(t),\tilde{\rho}_{k,n,t}\big)\big)\Big] \\
&\qquad - \mathbb{E}\Big[\big(\bar{\theta}_{i,j}(t) - \bar{\theta}^{(n)}_{i,j}(t)\big)\Big(\xi_j\big(\bar{\theta}_i(t)\big) - \xi_j\big(\theta^{(n)}_i(t)\big)\Big)\Big]\Big] \\
&= \frac{2}{k}\sum_{i=1}^{k}\sum_{j=1}^{d}\mathbb{E}\Big[\big(\bar{\theta}_{i,j}(t) - \bar{\theta}^{(n)}_{i,j}(t)\big)\big(b_j(\bar{\theta}_i(t),\rho_t) - \hat{b}_{n,j}\big(\theta^{(n)}_i(t),\tilde{\rho}_{k,n,t}\big)\big)\Big] \\
&\qquad - \frac{2}{k}\sum_{i=1}^{k}\mathbb{E}\Big[\big(\bar{\theta}_i(t) - \bar{\theta}^{(n)}_i(t)\big)\cdot\Big(\xi\big(\bar{\theta}_i(t)\big) - \xi\big(\theta^{(n)}_i(t)\big)\Big)\Big] \\
&\overset{(a)}{\le} \frac{2}{k}\sum_{i=1}^{k}\sum_{j=1}^{d}\mathbb{E}\Big[\big(\bar{\theta}_{i,j}(t) - \bar{\theta}^{(n)}_{i,j}(t)\big)\big(b_j(\bar{\theta}_i(t),\rho_t) - \hat{b}_{n,j}\big(\theta^{(n)}_i(t),\tilde{\rho}_{k,n,t}\big)\big)\Big] \\
&\qquad + \frac{2\beta_V}{k}\sum_{i=1}^{k}\mathbb{E}\Big[\big\|\bar{\theta}_i(t) - \bar{\theta}^{(n)}_i(t)\big\|^2\Big] \\
&\le \frac{1}{k}\sum_{i=1}^{k}\sum_{j=1}^{d}\mathbb{E}\Big[\big(\bar{\theta}_{i,j}(t) - \bar{\theta}^{(n)}_{i,j}(t)\big)^2 + \big(b_j(\bar{\theta}_i(t),\rho_t) - \hat{b}_{n,j}\big(\theta^{(n)}_i(t),\tilde{\rho}_{k,n,t}\big)^2\Big] \\
&\qquad + \frac{2\beta_V}{k}\sum_{i=1}^{k}\mathbb{E}\Big[\big\|\bar{\theta}_i(t) - \bar{\theta}^{(n)}_i(t)\big\|^2\Big] \\
&= (1+2\beta_V)\mathbb{E}\big[e_k(t)^2\big] + \frac{1}{k}\sum_{i=1}^{k}\sum_{j=1}^{d}\mathbb{E}\Big[\big(b_j(\bar{\theta}_i(t),\rho_t) - \hat{b}_{n,j}\big(\theta^{(n)}_i(t),\tilde{\rho}_{k,n,t}\big)^2\Big], \qquad (28)
\end{aligned}$$

where (a) used that β_V-semiconvexity of V implies that implies that for all $i \in [1:k]$,

$$\big(\bar{\theta}_i(t) - \bar{\theta}^{(n)}_i(t)\big)\cdot\Big(\xi\big(\bar{\theta}_i(t)\big) - \xi\big(\theta^{(n)}_i(t)\big)\Big) \ge -\beta_V\big\|\bar{\theta}_i(t) - \bar{\theta}^{(n)}_i(t)\big\|_2^2,$$

for all $\bar{\theta}_i(t)$, $\bar{\theta}^{(n)}_i(t)$, $\xi\big(\bar{\theta}_i(t)\big) \in \partial V\big(\bar{\theta}_i(t)\big)$ and $\xi\big(\bar{\theta}^{(n)}_i(t)\big) \in \partial V\big(\bar{\theta}^{(n)}_i(t)\big)$.

By Jensen's inequality,

$$\begin{aligned}
&\left|b_j(\bar{\theta}_i(t),\rho_t)-\hat{b}_{n,j}\big(\theta_i^{(n)}(t),\tilde{\rho}_{k,n,t}\big)\right|^2\\
&\le 3\left|b_j(\bar{\theta}_i(t),\rho_t)-b_j\big(\bar{\theta}_i(t),\bar{\rho}_{k,t}\big)\right|^2+3\left|b_j\big(\bar{\theta}_i(t),\bar{\rho}_{k,t}\big)-b_j\big(\theta_i^{(n)}(t),\tilde{\rho}_{k,n,t}\big)\right|^2\\
&\quad+3\left|b_j\big(\theta_i^{(n)}(t),\tilde{\rho}_{k,n,t}\big)-\hat{b}_{n,j}\big(\theta_i^{(n)}(t),\tilde{\rho}_{k,n,t}\big)\right|^2. \qquad (29)
\end{aligned}$$

For the first term, we obtain

$$\begin{aligned}
\left|b_j(\bar{\theta}_i(t),\rho_t)-b_j\big(\bar{\theta}_i(t),\bar{\rho}_{k,t}\big)\right| &= \left|\mathbb{E}_\mu\big[(\bar{R}'(g_{\rho_t})-\bar{R}'(g_{\bar{\rho}_{k,t}}))\partial_j\phi\big(\cdot,\bar{\theta}_i(t)\big)\big]\right|\\
&\le \mathbb{E}_\mu\Big[L_{\bar{R}',\tilde{\mathcal{G}}}\big|g_{\rho_t}-g_{\bar{\rho}_{k,t}}\big|\,\big|\partial_j\phi\big(\cdot,\bar{\theta}_i(t)\big|\Big]\\
&\le \big\|L_{\bar{R}',\tilde{\mathcal{G}}}\big\|_{2,\mu}\big\|(g_{\rho_t}-g_{\bar{\rho}_{k,t}})\partial_j\phi\big(\cdot,\bar{\theta}_i(t)\big\|_{2,\mu}.
\end{aligned}$$

Hence

$$\mathbb{E}\big[\,\big|b_j(\bar{\theta}_i(t),\rho_t)-b_j\big(\bar{\theta}_i(t),\bar{\rho}_{k,t}\big)\big|^2\,\big]\le\big\|L_{\bar{R}',\tilde{\mathcal{G}}}\big\|_{2,\mu}^2\mathbb{E}\big[\,\big\|(g_{\rho_t}-g_{\bar{\rho}_{k,t}})\partial_j\phi\big(\cdot,\bar{\theta}_i(t)\big\|_{2,\mu}^2\,\big].$$

The expectation in the RHS above can be written as

$$\begin{aligned}
\mathbb{E}\big[\,\big\|(g_{\rho_t}-g_{\bar{\rho}_{k,t}})\partial_j\phi\big(\cdot,\bar{\theta}_i(t)\big\|_{2,\mu}^2\,\big] &= \mathbb{E}\left[\int_{\mathcal{X}}(g_{\rho_t}(x)-g_{\bar{\rho}_{k,t}}(x))^2\big(\partial_j\phi(x,\bar{\theta}_i(t))\big)^2d\mu(x)\right]\\
&=\int_{\mathcal{X}}\mathbb{E}\left[(g_{\rho_t}(x)-g_{\bar{\rho}_{k,t}}(x))^2\big(\partial_j\phi(x,\bar{\theta}_i(t))\big)^2\right]d\mu(x).
\end{aligned}$$

Note that

$$\begin{aligned}
&\mathbb{E}\left[(g_{\rho_t}(x)-g_{\bar{\rho}_{k,t}}(x))^2\big(\partial_j\phi(x,\bar{\theta}_i(t))\big)^2\right]\\
&=\frac{1}{k^2}\mathbb{E}\left[\left(\sum_{l=1}^{k}\Big(\phi\big(x,\bar{\theta}_l(t)\big)-\mathbb{E}\big[\phi\big(x,\bar{\theta}(t)\big)\big]\Big)\partial_j\phi\big(x,\bar{\theta}_i(t)\big)\right)^2\right]\\
&=\frac{2}{k^2}\mathbb{E}\left[\left(\sum_{l\neq i}\Big(\phi\big(x,\bar{\theta}_l(t)\big)-\mathbb{E}\big[\phi\big(x,\bar{\theta}(t)\big)\big]\Big)\partial_j\phi\big(x,\bar{\theta}_i(t)\big)\right)^2\right]\\
&\qquad+\frac{2}{k^2}\mathbb{E}\left[\Big(\phi\big(x,\bar{\theta}_i(t)\big)-\mathbb{E}\big[\phi\big(x,\bar{\theta}(t)\big)\big]\Big)^2\Big(\partial_j\phi\big(x,\bar{\theta}_i(t)\big)\Big)^2\right]\\
&=\frac{2}{k^2}\mathbb{E}\left[\Big(\partial_j\phi\big(x,\bar{\theta}_i(t)\big)\Big)^2\mathbb{E}_{\rho_t^{\otimes k-1}}\left[\left(\sum_{l\neq i}\Big(\phi(x,\bar{\theta}_l(t))-\mathbb{E}\big[\phi\big(x,\bar{\theta}(t)\big)\big]\Big)\right)^2\right]\right]\\
&\qquad+\frac{2}{k^2}\mathbb{E}\left[\Big(\phi\big(x,\bar{\theta}_i(t)\big)-\mathbb{E}\big[\phi(y,\bar{\theta}(t))\big]\Big)^2\Big(\partial_j\phi\big(x,\bar{\theta}_i(t)\big)\Big)^2\right].
\end{aligned}$$

Similar to (24), we have

$$\mathbb{E}_{\rho_t^{\otimes k-1}}\left[\left(\sum_{l\neq i}\Big(\phi(x,\bar{\theta}_l(t))-\mathbb{E}\big[\phi\big(x,\bar{\theta}(t)\big)\big]\Big)\right)^2\right]=(k-1)\mathsf{Var}_{\rho_t}\big[\phi\big(x,\bar{\theta}(t)\big)\big].$$

Hence

$$\begin{aligned}
&\mathbb{E}\left[(g_{\rho_t}(x)-g_{\bar\rho_{k,t}}(x))^2\big(\partial_j\phi(x,\bar\theta_i(t))\big)^2\right]\\
&=\frac{2(k-1)}{k^2}\mathsf{Var}_{\rho_t}\big[\phi\big(x,\bar\theta(t)\big)\big]\mathbb{E}_{\rho_t}\left[\big(\partial_j\phi\big(x,\bar\theta(t)\big)\big)^2\right]\\
&\qquad+\frac{2}{k^2}\mathbb{E}\left[\Big(\phi\big(x,\bar\theta_i(t)\big)-\mathbb{E}\big[\phi\big(x,\bar\theta(t)\big)\big]\Big)^2\Big(\partial_j\phi\big(x,\bar\theta_i(t)\big)\Big)^2\right].
\end{aligned}$$

Since the Lipschitz constant of $\phi(x,\cdot):\Theta\to\mathbb{R}$ is bounded by $C_{\mathsf{L}}\|x\|$, we have

$$\Big|\phi\big(x,\theta\big)-\phi\big(x,\tilde\theta\big)\Big|\le C_{\mathsf{L}}\|x\|\,\big\|\theta-\tilde\theta\big\|.$$

So, with $\big(\bar\theta(t),\hat\theta(t)\big)\sim\rho_t^{\otimes 2}$,

$$\begin{aligned}
\mathsf{Var}_{\rho_t}\big[\phi\big(x,\bar\theta(t)\big)\big]&\le\mathbb{E}_{\rho_t^{\otimes 2}}\left[\big(\phi\big(x,\bar\theta(t)\big)-\phi\big(x,\hat\theta(t)\big)\big)^2\right]\\
&\le C_{\mathsf{L}}^2\,\|x\|^2\,\mathbb{E}_{\rho_t^{\otimes 2}}\left[\big\|\bar\theta(t)-\hat\theta(t)\big\|^2\right]\\
&\le 2C_{\mathsf{L}}^2\,\|x\|^2\,\mathbb{E}_{\rho_t^{\otimes 2}}\left[\big\|\bar\theta(t)\big\|^2+\big\|\hat\theta(t)\big\|^2\right]\\
&=4C_{\mathsf{L}}^2\,C_{\mathsf{M}}\,\|x\|^2.
\end{aligned}$$

Summing over $j\in[1:d]$, we obtain

$$\begin{aligned}
\sum_{j=1}^d\mathbb{E}\left[(g_{\rho_t}(x)-g_{\bar\rho_{k,t}}(x))^2\big(\partial_j\phi(x,\bar\theta_i(t))\big)^2\right]&=\mathbb{E}\left[(g_{\rho_t}(x)-g_{\bar\rho_{k,t}}(x))^2\,\big\|\nabla_\theta\phi(x,\bar\theta_i(t))\big)\big\|_2^2\right]\\
&\le\frac{8\,C_{\mathsf{L}}^4\,(k-1)}{k^2}\,\|x\|^4\,M_2(\rho_t)+\frac{8\,C_{\mathsf{L}}^4}{k^2}\,\|x\|^4\,M_2(\rho_t)\\
&=\frac{8C_{\mathsf{L}}^4C_{\mathsf{M}}}{k}\,\|x\|^4.
\end{aligned}$$

Hence

$$\begin{aligned}
\sum_{j=1}^d\mathbb{E}\big[\,\big|b_j(\bar\theta_i(t),\rho_t)-b_j\big(\bar\theta_i(t),\bar\rho_{k,t}\big)\big|^2\,\big]&\le\big\|L_{\bar R',\tilde{\mathcal{G}}}\big\|_{2,\mu}^2\sum_{j=1}^d\mathbb{E}\big[\,\big\|(g_{\rho_t}-g_{\bar\rho_{k,t}})\partial_j\phi\big(\cdot,\bar\theta_i(t)\big\|_{2,\mu}^2\,\big]\\
&\le\frac{8\,C_{\mathsf{L}}^4}{k}M_2(\rho_t)\big\|L_{\bar R',\tilde{\mathcal{G}}}\big\|_{2,\mu}^2\int_{\mathcal{X}}\|x\|^4\,d\mu(x)\\
&\le\frac{8\,C_{\mathsf{L}}^4C_{\mathsf{M}}}{k}\big\|L_{\bar R',\tilde{\mathcal{G}}}\big\|_{2,\mu}^2M_4(\mu).
\end{aligned}\tag{30}$$

For the last term in (29),

$$\begin{aligned}
&\sum_{j=1}^d\mathbb{E}\left[\Big|b_j\big(\theta_i^{(n)}(t),\tilde\rho_{k,n,t}\big)-\hat b_{n,j}\big(\theta_i^{(n)}(t),\tilde\rho_{k,n,t}\big)\Big|^2\right]\\
&=\sum_{j=1}^d\mathbb{E}\left[\Big|\mathbb{E}_{\hat\mu_n}\Big[\bar R'\big(g_{\tilde\rho_{k,n,t}}\big)\partial_j\phi\big(\cdot,\theta_i^{(n)}(t)\big)\Big]-\mathbb{E}_{\mu}\Big[\bar R'\big(g_{\tilde\rho_{k,n,t}}\big)\partial_j\phi\big(\cdot,\theta_i^{(n)}(t)\big)\Big]\Big|^2\right]
\end{aligned}$$

$$= \frac{1}{n}\sum_{j=1}^{d}\mathbb{E}\left[\left|\bar{R}'\big(g_{\tilde{\rho}_{k,n,t}}\big)\partial_j\phi\big(\cdot,\theta_i^{(n)}(t)\big) - \mathbb{E}_\mu\Big[\bar{R}'\big(g_{\tilde{\rho}_{k,n,t}}\big)\partial_j\phi\big(\cdot,\theta_i^{(n)}(t)\big)\Big]\right|^2\right] \tag{31}$$

$$\leq \frac{1}{n}\sum_{j=1}^{d}\mathbb{E}\left[\Big(\bar{R}'\big(g_{\tilde{\rho}_{k,n,t}}\big)\partial_j\phi\big(\cdot,\theta_i^{(n)}(t)\big)\Big)^2\right]$$

$$\leq \frac{1}{n}\sum_{j=1}^{d}\mathbb{E}\left[\Big(G_{\bar{R}',\tilde{\mathcal{G}}}\partial_j\phi\big(\cdot,\theta_i^{(n)}(t)\big)\Big)^2\right]$$

$$= \frac{1}{n}\mathbb{E}\left[G^2_{\bar{R}',\tilde{\mathcal{G}}}\left\|\nabla_\theta\phi\big(X,\theta_i^{(n)}(t)\big)\right\|_2^2\right]$$

$$\leq \frac{1}{n}\mathbb{E}\left[G^2_{\bar{R}',\tilde{\mathcal{G}}}\left\|\nabla_\theta\phi\big(X,\theta_i^{(n)}(t)\big)\right\|_2^2\right]$$

$$\leq \frac{C_{\mathsf{L}}^2}{n}\mathbb{E}\left[G^2_{\bar{R}',\tilde{\mathcal{G}}}\|X\|_2^2\right], \tag{32}$$

where (31) follows similar to (24). Finally, considering the second term in (29),

$$\sum_{j=1}^{d}\left|b_j\big(\bar{\theta}_i(t),\bar{\rho}_{k,t}\big) - b_j\big(\theta_i^{(n)}(t),\tilde{\rho}_{k,n,t}\big)\right|^2$$

$$= \sum_{j=1}^{d}\left|\mathbb{E}_\mu\big[\bar{R}'(g_{\bar{\rho}_{k,t}})\partial_j\phi\big(\cdot,\bar{\theta}_i(t)\big)\big] - \mathbb{E}_\mu\big[\bar{R}'(g_{\tilde{\rho}_{k,n,t}})\partial_j\phi\big(\cdot,\theta_i^{(n)}(t)\big)\big]\right|^2$$

$$\leq \sum_{j=1}^{d}\Bigg(2\left|\mathbb{E}_\mu\big[\bar{R}'(g_{\bar{\rho}_{k,t}})\partial_j\phi\big(\cdot,\bar{\theta}_i(t)\big)\big] - \mathbb{E}_\mu\big[\bar{R}'(g_{\bar{\rho}_{k,t}})\partial_j\phi\big(\cdot,\theta_i^{(n)}(t)\big)\big]\right|^2$$

$$+ 2\left|\mathbb{E}_\mu\big[\bar{R}'(g_{\bar{\rho}_{k,t}})\partial_j\phi\big(\cdot,\theta_i^{(n)}(t)\big)\big] - \mathbb{E}_\mu\big[\bar{R}'(g_{\tilde{\rho}_{k,n,t}})\partial_j\phi\big(\cdot,\theta_i^{(n)}(t)\big)\big]\right|^2\Bigg)$$

$$\leq \sum_{j=1}^{d}\Bigg(2\left\|G_{\bar{R}',\tilde{\mathcal{G}}}\right\|_{2,\mu}^2\left\|\partial_j\phi\big(\cdot,\bar{\theta}_i(t)\big) - \partial_j\phi\big(\cdot,\theta_i^{(n)}(t)\big)\right\|_{2,\mu}^2$$

$$+ 2\Big(\mathbb{E}_\mu\big[\big(\bar{R}'(g_{\bar{\rho}_{k,t}}) - \bar{R}'(g_{\tilde{\rho}_{k,n,t}})\big)\partial_j\phi\big(\cdot,\theta_i^{(n)}(t)\big)\big]\Big)^2\Bigg)$$

$$\leq \sum_{j=1}^{d}\Bigg(2\left\|G_{\bar{R}',\tilde{\mathcal{G}}}\right\|_{2,\mu}^2\left\|\partial_j\phi\big(\cdot,\bar{\theta}_i(t)\big) - \partial_j\phi\big(\cdot,\theta_i^{(n)}(t)\big)\right\|_{2,\mu}^2$$

$$+ 2\Big(\mathbb{E}_\mu\big[\big|L_{\bar{R}',\tilde{\mathcal{G}}}\big|\,\big|g_{\bar{\rho}_{k,t}} - g_{\tilde{\rho}_{k,n,t}}\big|\,\big|\partial_j\phi\big(\cdot,\theta_i^{(n)}(t)\big)\big|\big]\Big)^2\Bigg)$$

$$\leq \sum_{j=1}^{d}\Bigg(2\left\|G_{\bar{R}',\tilde{\mathcal{G}}}\right\|_{2,\mu}^2\left\|\partial_j\phi\big(\cdot,\bar{\theta}_i(t)\big) - \partial_j\phi\big(\cdot,\theta_i^{(n)}(t)\big)\right\|_{2,\mu}^2$$

$$+ 2\left\|L_{\bar{R}',\tilde{\mathcal{G}}}\big|\partial_j\phi\big(\cdot,\theta_i^{(n)}(t)\big)\big|\right\|_{2,\mu}^2\left\|g_{\bar{\rho}_{k,t}} - g_{\tilde{\rho}_{k,n,t}}\right\|_{2,\mu}^2\Bigg)$$

$$= 2\left\|G_{\bar{R}',\tilde{\mathcal{G}}}\right\|_{2,\mu}^2\left\|\nabla\phi\big(\cdot,\bar{\theta}_i(t)\big) - \nabla\phi\big(\cdot,\theta_i^{(n)}(t)\big)\right\|_{2,\mu}^2$$

$$+ 2\left\|L_{\bar{R}',\tilde{\mathcal{G}}}\left\|\nabla\phi\big(\cdot,\theta_i^{(n)}(t)\big)\right\|_2\right\|_{2,\mu}^2\left\|g_{\bar{\rho}_{k,t}} - g_{\tilde{\rho}_{k,n,t}}\right\|_{2,\mu}^2$$

$$\leq 2C_{\mathsf{L}}^2 M_2(\mu) \left\| G_{\bar{R}',\tilde{\mathcal{G}}} \right\|_{2,\mu}^2 \left\| \bar{\theta}_i(t) - \theta_i^{(n)}(t) \right\|_2^2 + 2C_{\mathsf{L}}^2 \left\| L_{\bar{R}',\tilde{\mathcal{G}}} \left\| X \right\|_2 \right\|_{2,\mu}^2 \left\| g_{\bar{\rho}_{k,t}} - g_{\tilde{\rho}_{k,n,t}} \right\|_{2,\mu}^2$$

$$\leq 2C_{\mathsf{L}}^2 M_2(\mu) \left\| G_{\bar{R}',\tilde{\mathcal{G}}} \right\|_{2,\mu}^2 \left\| \bar{\theta}_i(t) - \theta_i^{(n)}(t) \right\|_2^2 + 2C_{\mathsf{L}}^4 \left\| L_{\bar{R}',\tilde{\mathcal{G}}} \left\| X \right\|_2 \right\|_{2,\mu}^2 e_k^2 M_2(\mu), \tag{33}$$

where the final inequality is because using Jensen's inequality, we have

$$\begin{aligned}
\left\| g_{\bar{\rho}_{k,t}} - g_{\tilde{\rho}_{k,n,t}} \right\|_{2,\mu}^2 &= \left\| \frac{1}{k} \sum_{i=1}^{k} \phi(\cdot, \bar{\theta}_i(t)) - \phi\big(\cdot, \theta_i^{(n)}(t)\big) \right\|_{2,\mu}^2 \\
&\leq \frac{1}{k} \sum_{i=1}^{k} \left\| \phi(\cdot, \bar{\theta}_i(t)) - \phi\big(\cdot, \theta_i^{(n)}(t)\big) \right\|_{2,\mu}^2 \\
&\leq C_{\mathsf{L}}^2 \left(\int_{\mathcal{X}} \|x\|_2^2 \, d\mu(x) \right) \frac{1}{k} \sum_{i=1}^{k} \left\| \bar{\theta}_i(t)) - \phi\big(\cdot, \theta_i^{(n)} \right\|_2^2 \\
&= C_{\mathsf{L}}^2 e_k^2 M_2(\mu).
\end{aligned}$$

Taking expectation in (33),

$$\begin{aligned}
&\frac{1}{k} \sum_{i=1}^{k} \sum_{j=1}^{d} \mathbb{E}\left[\left| b_j\big(\bar{\theta}_i(t), \bar{\rho}_{k,t}\big) - b_j\big(\theta_i^{(n)}(t), \tilde{\rho}_{k,n,t}\big) \right|^2 \right] \\
&\leq \big(2C_{\mathsf{L}}^2 \, M_2(\mu) \left\| G_{\bar{R}',\tilde{\mathcal{G}}} \right\|_{2,\mu}^2 + 2C_{\mathsf{L}}^4 \left\| L_{\bar{R}',\tilde{\mathcal{G}}} \left\| X \right\|_2 \right\|_{2,\mu}^2 M_2(\mu)\big) \mathbb{E}\big[e_k^2\big].
\end{aligned}$$

From (28), (29), (30), (32) and the above equation,

$$\begin{aligned}
\frac{d}{dt} \mathbb{E}\big[e_k(t)^2\big] \leq \Bigg(1 + 2\beta_V + 6C_{\mathsf{L}}^2 \, M_2(\mu) \left\| G_{\bar{R}',\tilde{\mathcal{G}}} \right\|_{2,\mu}^2 + 6C_{\mathsf{L}}^4 \left\| L_{\bar{R}',\tilde{\mathcal{G}}} \left\| X \right\|_2 \right\|_{2,\mu}^2 M_2(\mu) \Bigg) \mathbb{E}\big[e_k(t)^2\big] \\
+ \frac{24\, C_{\mathsf{L}}^4 C_{\mathsf{M}}}{k} \left\| L_{\bar{R}',\tilde{\mathcal{G}}} \right\|_{2,\mu}^2 M_4(\mu) + \frac{3C_{\mathsf{L}}^2}{n} \mathbb{E}_\mu \Big[G_{\bar{R}',\tilde{\mathcal{G}}}^2 \left\| X \right\|_2^2 \Big].
\end{aligned}$$

Since $\bar{\boldsymbol{\theta}}(0) = \boldsymbol{\theta}^{(n)}(0)$ by assumption of perfect coupling of particles at time $t = 0$, $e_k(0) = 0$ almost surely. Then, by a straightforward application of Gronwall's lemma as given below with $u(t) = \mathbb{E}\big[e_k^2(t)\big]$ and $d(t)$ a constant, we obtain (27).

Lemma 5 (Gronwall's lemma) *Suppose* $u : [0, T] \to [0, \infty)$ *satisfies*

$$\frac{du(t)}{dt} \leq a u(t) + d(t),$$

for $a \geq 0$*, then*

$$u(t) \leq e^{at} \left(u(0) + \int_0^t e^{-as} d(s) ds \right).$$

In particular, if $d(t) \leq c$ *for some constant* c *independent of* t*, then* $u(t) \leq e^{at} u(0) + \frac{c}{a}(e^{at} - 1)$.

C. Wasserstein Gradient Flow Convergence in Mean Field Limit

IV. Neural Estimation of Statistical Divergences

Let $\mu, \nu \in \mathcal{P}(\mathcal{X})$ for some $\mathcal{X} \subseteq \mathbb{R}^m$. Here, we consider neural estimation of statistical divergences between μ, ν, which are unknown, based on samples X^n and Y^n generated i.i.d. according to the distributions $\mu^{\otimes n}$ and $\nu^{\otimes n}$, respectively. Specifically, we are interested in divergences that admit a variational form:

$$\mathsf{D}(\mu, \nu) = \sup_{f \in \mathcal{F}} \mathbb{E}_\mu[f] - \mathbb{E}_\nu[h \circ f], \tag{34}$$

where $h : \mathbb{R} \to \mathbb{R}$ denotes some convex function and $\mathcal{F}$ is a class of functions that both depends on the specific divergence. With $\tilde{\mathcal{P}}_2(\Theta) = \tilde{\mathcal{P}}_{2,\mathsf{Leb},H}(\Theta, C_{\mathsf{M}})$, the mean field optimization problem of interest is

$$\bar{\mathsf{D}}\big(\mu, \nu, \lambda, \tau, \tilde{\mathcal{P}}_2(\Theta)\big) \coloneqq \sup_{\rho \in \tilde{\mathcal{P}}_2(\Theta)} \mathbb{E}_\mu[g_\rho] - \mathbb{E}_\nu[h \circ g_\rho] - \lambda \int V d\rho - \tau H(\rho). \tag{35}$$

For simplicity, we will also assume that $\nabla_\theta V(\theta)$ exists for all $\theta \in \mathsf{int}(\Theta)$. This problem is equivalent to (2) with $J\big(\rho, \lambda, \tau\big) = \mathbb{E}_\nu[h \circ g_\rho] - \mathbb{E}_\mu[g_\rho] + \lambda \int V d\rho + \tau U(\rho)$. When $\lambda = \tau = 0$, $\bar{\mathsf{D}}\big(\mu, \nu, \lambda, \tau, \tilde{\mathcal{P}}_2(\Theta)\big)$ is a discrepancy measure between μ and ν, referred to as a neural net distance.

For this problem, we will consider the following random finite particle neural estimate:

$$\hat{\mathsf{D}}_{k,n}\big(\hat{\mu}_n, \hat{\nu}_n, \lambda, \hat{\mathcal{P}}_k(\Theta)\big) \coloneqq \sup_{\rho \in \hat{\mathcal{P}}_k(\Theta)} \mathbb{E}_{\hat{\mu}_n}[g_\rho] - \mathbb{E}_{\hat{\nu}_n}[h \circ g_\rho] - \lambda \int V d\rho, \tag{36}$$

where $\hat{\mu}_n \coloneqq \left(\sum_{i=1}^n \delta_{X_i}\right)/n$, $\hat{\nu}_n \coloneqq \left(\sum_{i=1}^n \delta_{Y_i}\right)/n$. The framework in (35) and (36) is general enough to capture neural estimation of f-divergences, integral probability metrics, and certain optimal transport metrics and their regularized versions, by noting that these quantities admit the variational form in (34) with different choices of h and $\mathcal{F}$. If the function class induced by $\tilde{\mathcal{P}}_2(\Theta)$ is contained in $\mathcal{F}$ and can approximate the optimizer of the RHS of (34) with arbitrary accuracy (say, in uniform norm), then it can be seen that the approximation error

$$\mathsf{Approx.\ error} \coloneqq \left|\bar{\mathsf{D}}_h\big(\mu, \nu, \lambda, \tau, \tilde{\mathcal{P}}_2(\Theta)\big) - \mathsf{D}_{h,\mathcal{F}}(\mu, \nu)\right|, \tag{37}$$

can be made arbitrary small for λ and τ sufficiently small. Since there is extensive literature on neural approximation error bounds for standard function classes (e.g., polynomials, Barron, Hölder, Besov), we do not delve into this aspect much and treat the approximation error as given. Henceforth, we will focus on the more general problem in (2).

A. Optimization Error for Kullback-Leibler Divergence

The Kullback-Leibler (KL) divergence between distributions $\mu, \nu \in \mathcal{P}(\mathcal{X})$ is

$$\mathsf{D}_{\mathsf{KL}}(\mu\|\nu) \coloneqq \begin{cases} \int_{\mathcal{X}} \log \frac{d\mu}{d\nu} d\mu, & \mu \ll \nu, \\ \infty, & \text{otherwise.} \end{cases} \tag{38}$$

It admits the variational expression

$$\mathsf{D}_{\mathsf{KL}}(\mu\|\nu) = \sup_{f \in \mathcal{F}} \mathbb{E}_{\mu}[f] - \mathbb{E}_{\nu}[e^f] + 1, \tag{39}$$

where the supremum is over all measurable functions such that the second expectation is finite. The relevant minimization problem (4) corresponding to the finite particle neural estimate is

$$\tilde{\mathsf{D}}^{(\mathsf{KL})}_{k,n}\big(\hat{\mu}_n, \hat{\nu}_n, \lambda, \hat{\mathcal{P}}_k(\Theta)\big) \coloneqq - \inf_{\rho \in \hat{\mathcal{P}}_k(\Theta)} \tilde{\mathsf{D}}^{(\mathsf{KL})}_{k,n}\big(\hat{\mu}_n, \hat{\nu}_n, \lambda, \rho\big),$$
$$\tilde{\mathsf{D}}^{(\mathsf{KL})}_{k,n}\big(\hat{\mu}_n, \hat{\nu}_n, \lambda, \rho\big) \coloneqq \mathbb{E}_{\hat{\nu}_n}[e^{g_\rho}] - \mathbb{E}_{\hat{\mu}_n}[g_\rho] - 1 + \lambda \int V d\rho,$$

Let $\tilde{\mathcal{P}}_{2,\mathsf{Leb},H}(\Theta, C_{\mathsf{M}})$ denote the subset of $\mathcal{P}_2(\Theta)$ with Lebesgue densities, finite entropy, finite Fisher information and $M_2(\rho) \leq C_{\mathsf{M}}$. The mean field optimization problem of interest is

$$\bar{\mathsf{D}}_{\mathsf{KL}}\big(\mu, \nu, \lambda, \tau, \tilde{\mathcal{P}}_2(\Theta)\big) \coloneqq - \inf_{\rho \in \tilde{\mathcal{P}}_{2,\mathsf{Leb},H}(\Theta, C_{\mathsf{M}})} \bar{\mathsf{D}}_{\mathsf{KL}}\big(\mu, \nu, \lambda, \tau, \rho\big), \tag{40}$$
$$\bar{\mathsf{D}}_{\mathsf{KL}}\big(\mu, \nu, \lambda, \tau, \rho\big) \coloneqq \mathbb{E}_{\nu}[e^{g_\rho}] - \mathbb{E}_{\mu}[g_\rho] - 1 + \lambda \int V d\rho + \tau H(\rho).$$

Let ρ_t be the Wasserstein gradient flow at time t according to the mean field objective $\bar{\mathsf{D}}_{\mathsf{KL}}\big(\mu, \nu, \lambda, \tau, \rho\big)$, i.e., with ρ_t denoting the law of $\theta(t)$ satisfies

$$\theta'(t) \coloneqq \frac{d}{dt}\theta(t) = \mathbb{E}_{\mu}\big[\nabla_\theta \phi\big(\cdot, \theta(t)\big)\big] - \mathbb{E}_{\nu}\big[e^{g_{\rho_t}} \nabla_\theta \phi\big(\cdot, \theta(t)\big)\big] - \lambda \nabla_\theta V\big(\theta(t)\big) - \tau \frac{\nabla_\theta \rho_t}{\rho_t}\big(\theta(t)\big).$$

The same evolution of ρ_t is also produced by the Langevin diffusion:

$$d\theta(t) \coloneqq \mathbb{E}_{\mu}\big[\nabla_\theta \phi\big(\cdot, \theta(t)\big)\big]dt - \mathbb{E}_{\nu}\big[e^{g_{\rho_t}} \nabla_\theta \phi\big(\cdot, \theta(t)\big)\big]dt - \lambda \nabla_\theta V\big(\bar{\theta}_i(t)\big)dt + \sqrt{2\tau}\, dB(t).$$

Let $\tilde{\rho}_{k,n,t}$ denote the empirical probability measure of $\boldsymbol{\theta}^{(n)}(t)$ at time t corresponding to particle gradient flow starting from $\boldsymbol{\theta}^{(n)}(0) = \bar{\boldsymbol{\theta}}(0)$. The relevant random finite particle evolution is governed by the Langevin diffusion:

$$d\theta^{(n)}_i(t) = \mathbb{E}_{\hat{\mu}_n}\left[\nabla_\theta \phi\big(\cdot, \theta^{(n)}_i(t)\big)\right] dt - \mathbb{E}_{\hat{\nu}_n}\left[e^{g_{\tilde{\rho}_{k,n,t}}} \nabla_\theta \phi\big(\cdot, \theta^{(n)}_i(t)\big)\right] dt - \lambda \nabla_\theta V\big(\theta^{(n)}_i(t)\big)dt$$
$$+ \sqrt{2\tau}\, dB_i(t). \tag{41}$$

Defining

$$\hat{\mathsf{D}}^{(\mathsf{KL})}_{k,n,t}(\hat{\mu}_n, \hat{\nu}_n) \coloneqq \mathbb{E}_{\hat{\mu}_n}[g_{\tilde{\rho}_{k,n,t}}] - \mathbb{E}_{\hat{\nu}_n}[e^{g_{\tilde{\rho}_{k,n,t}}}] + 1, \tag{42}$$

$$\check{\mathsf{D}}_t^{(\mathsf{KL})}(\mu,\nu) \coloneqq \mathbb{E}_\mu[g_{\rho_t}] - \mathbb{E}_\nu[e^{g_{\rho_t}}] + 1,$$

we next obtain a bound on their difference at time t.

Theorem 6 (Optimization error bound for KL) *Let $\mu,\nu \in \mathcal{P}_2(\mathcal{X})$ and $\tilde{\mathcal{P}}_2(\Theta) = \tilde{\mathcal{P}}_{2,\mathsf{Leb},H}(\Theta, C_{\mathsf{M}})$. Suppose Assumption 2 and the assumptions in Theorem 3 hold with $\bar{R}(g) = g$ and $\bar{R}(g) = e^g$ with respective to measures μ and ν, respectively. Then*

$$\begin{aligned}
&\mathbb{E}\left[\left|\check{\mathsf{D}}_t^{(\mathsf{KL})}(\mu,\nu) - \hat{\mathsf{D}}_{k,n,t}^{(\mathsf{KL})}(\hat{\mu}_n,\hat{\nu}_n)\right|\right] \\
&\le \left(\left\|\tilde{G}\right\|_{2,\mu}\sqrt{M_2(\mu)} + \left\|e^{\tilde{G}}\right\|_{2,\nu}\sqrt{M_2(\nu)}\right)\left(\sqrt{\frac{C_{\mathsf{M}}}{k}} + C_{\mathsf{L}}\left(\frac{\sqrt{\bar{c}_{\mu,\nu,\tilde{G}}}}{\sqrt{n}} + \frac{\sqrt{\tilde{c}_{\mu,\nu,G}}}{\sqrt{k}}\right)e^{t\,c_{\mu,\nu,\tilde{G}}}\right) \\
&\quad + \frac{C_{\mathsf{D}}}{\sqrt{n}}\left(E\big(\tilde{\mathcal{G}},\tilde{G}\big)\left\|\tilde{G}\right\|_{2,\mu} + E\big(e^{\tilde{\mathcal{G}}},e^{\tilde{G}}\big)\left\|e^{\tilde{G}}\right\|_{2,\nu}\right),
\end{aligned} \tag{43}$$

where

$$c_{\mu,\nu,\tilde{G}} \coloneqq 0.5\left(1 + 2\beta_V + 9C_{\mathsf{L}}^2\Big(M_2(\mu) + \left\|e^{\tilde{G}}\right\|_{2,\nu}^2 M_2(\nu)\Big) + 9C_{\mathsf{L}}^4\left\|e^{2\tilde{G}}\right\|_{2,\nu}\sqrt{M_4(\nu)M_2(\nu)}\right), \tag{44a}$$

$$\tilde{c}_{\mu,\nu,\tilde{G}} \coloneqq \frac{12\,C_{\mathsf{L}}^4 C_{\mathsf{M}}\left\|e^{\tilde{G}}\right\|_{2,\nu}^2 M_4(\nu)}{c_{\mu,\nu,\tilde{G}}}, \tag{44b}$$

$$\bar{c}_{\mu,\nu,\tilde{G}} \coloneqq \frac{3C_{\mathsf{L}}^2 M_2(\mu) + 3C_{\mathsf{L}}^2\left\|e^{2\tilde{G}}\right\|_{2,\nu}\sqrt{M_4(\nu)}}{c_{\mu,\nu,\tilde{G}}}. \tag{44c}$$

Proof. Since the proof is similar to that of Theorem 3, we will only highlight the differences. By triangle inequality,

$$\begin{aligned}
\left|\check{\mathsf{D}}_t^{(\mathsf{KL})}(\mu,\nu) - \hat{\mathsf{D}}_{k,n,t}^{(\mathsf{KL})}(\hat{\mu}_n,\hat{\nu}_n)\right| &= \left|\mathbb{E}_\mu[g_{\rho_t}] - \mathbb{E}_\nu[e^{g_{\rho_t}}] - \mathbb{E}_{\hat{\mu}_n}[g_{\tilde{\rho}_{k,n,t}} + \mathbb{E}_{\hat{\nu}_n}[e^{g_{\tilde{\rho}_{k,n,t}}}]]\right| \\
&\le \left|\mathbb{E}_\nu[e^{g_{\rho_t}}] - \mathbb{E}_{\hat{\nu}_n}[e^{g_{\tilde{\rho}_{k,n,t}}}]\right| + \left|\mathbb{E}_\mu[g_{\rho_t}] - \mathbb{E}_{\hat{\mu}_n}[g_{\tilde{\rho}_{k,n,t}}]\right|.
\end{aligned} \tag{45}$$

The first term can be bounded as follows:

$$\left|\mathbb{E}_\nu[e^{g_{\rho_t}}] - \mathbb{E}_{\hat{\nu}_n}[e^{g_{\tilde{\rho}_{k,n,t}}}]\right| \le \left|\mathbb{E}_\nu[e^{g_{\tilde{\rho}_{k,n,t}}}] - \mathbb{E}_{\hat{\nu}_n}[e^{g_{\tilde{\rho}_{k,n,t}}}]\right| + \left|\mathbb{E}_\nu[e^{g_{\rho_t}}] - \mathbb{E}_\nu[e^{g_{\tilde{\rho}_{k,n,t}}}]\right|. \tag{46}$$

Similar to (19),

$$\mathbb{E}\left[\left|\mathbb{E}_\nu[e^{g_{\tilde{\rho}_{k,n,t}}}] - \mathbb{E}_{\hat{\nu}_n}[e^{g_{\tilde{\rho}_{k,n,t}}}]\right|\right] \le \mathbb{E}\left[\left\|\hat{\nu}_n - \nu\right\|_{e^{\tilde{\mathcal{G}}}}\right] \le \frac{C_{\mathsf{D}}E\big(e^{\tilde{\mathcal{G}}},e^{\tilde{G}}\big)\left\|e^{\tilde{G}}\right\|_{2,\nu}}{\sqrt{n}},$$

where $\tilde{\mathcal{G}}$ is as defined in (13) with envelope $\tilde{G}$ and $E\big(\mathcal{F},F\big)$ is as defined in (14).

For the second term in (46), an application of Cauchy-Schwarz inequality yields

$$\left|\mathbb{E}_\nu[e^{g_{\rho_t}}] - \mathbb{E}_\nu[e^{g_{\tilde{\rho}_{k,n,t}}}]\right| \le \left\|\left(e^{g_{\rho_t}} \vee e^{g_{\tilde{\rho}_{k,n,t}}}\right)\left|g_{\rho_t} - g_{\tilde{\rho}_{k,n,t}}\right|\right\|_{2,\nu}$$

$$\leq \|e^{g_{\rho_t}} \vee e^{g_{\bar{\rho}_{k,n,t}}}\|_{2,\nu} \left\|g_{\rho_t} - g_{\bar{\rho}_{k,n,t}}\right\|_{2,\nu}$$
$$\leq \left\|e^{\tilde{G}}\right\|_{2,\nu} \left\|g_{\rho_t} - g_{\bar{\rho}_{k,n,t}}\right\|_{2,\nu}.$$

Hence

$$\mathbb{E}\left[\left|\mathbb{E}_{\nu}[e^{g_{\rho_t}}] - \mathbb{E}_{\hat{\nu}_n}[e^{g_{\bar{\rho}_{k,n,t}}}]\right|\right] \leq \frac{C_{\mathsf{D}} E\big(e^{\tilde{\mathcal{G}}}, e^{\tilde{G}}\big)\left\|e^{\tilde{G}}\right\|_{2,\nu}}{\sqrt{n}} + \left\|e^{\tilde{G}}\right\|_{2,\nu} \mathbb{E}\left[\left\|g_{\rho_t} - g_{\bar{\rho}_{k,n,t}}\right\|_{2,\nu}\right]. \tag{47}$$

For bounding the last term in (47), we will use a propagation of chaos inspired technique as in Theorem 3. To this end, let $\bar{\boldsymbol{\theta}}(t) = \big(\bar{\theta}_1(t), \ldots, \bar{\theta}_k(t)\big) \sim \rho_t^{\otimes k}$. Consider that the particles $\{\bar{\theta}_i(t)\}_{i=1}^k$ each evolve independently at time t starting from $\bar{\boldsymbol{\theta}}(0) \sim \rho_0^{\otimes k}$ according to the mean field dynamics:

$$d\bar{\theta}_i(t) = \mathbb{E}_{\mu}\big[\nabla_\theta \phi\big(\cdot, \bar{\theta}_i(t)\big)\big]dt - \mathbb{E}_{\nu}\big[e^{g_{\rho_t}} \nabla_\theta \phi\big(\cdot, \bar{\theta}_i(t)\big)\big]dt - \lambda \nabla_\theta V\big(\bar{\theta}_i(t)\big)dt + \sqrt{2\tau}\, dB_i(t). \tag{48}$$

Let $\bar{\rho}_{k,t}(\bar{\boldsymbol{\theta}}(t)) = \big(\sum_{i=1}^k \delta_{\bar{\theta}_i(t)}\big)/k$ denote the empirical distribution of $\bar{\boldsymbol{\theta}}(t)$. Using triangle inequality,

$$\left\|g_{\rho_t} - g_{\bar{\rho}_{k,n,t}}\right\|_{2,\nu} \leq \left\|g_{\rho_t} - g_{\bar{\rho}_{k,t}}\right\|_{2,\nu} + \left\|g_{\bar{\rho}_{k,t}} - g_{\bar{\rho}_{k,n,t}}\right\|_{2,\nu}. \tag{49}$$

Since $\{\bar{\theta}_i\}_{i=1}^k$ are i.i.d., we obtain similar to (23)

$$\mathbb{E}_{\rho_t^{\otimes k}}\left[\left\|g_{\rho_t} - g_{\bar{\rho}_{k,t}}\right\|_{2,\nu}^2\right] \leq \frac{C_{\mathsf{M}} M_2(\nu)}{k}, \tag{50}$$

For the second term in the RHS of (49), we have similar to (26) that

$$\mathbb{E}\left[\left\|g_{\bar{\rho}_{k,t}} - g_{\bar{\rho}_{k,n,t}}\right\|_{2,\nu}\right] \leq \sqrt{\mathbb{E}\left[\left\|g_{\bar{\rho}_{k,t}} - g_{\bar{\rho}_{k,n,t}}\right\|_{2,\nu}^2\right]} \leq C_{\mathsf{L}} \sqrt{M_2(\nu)\mathbb{E}\big[e_k^2(t)\big]}, \tag{51}$$

where $e_k^2(t)$ is as defined in (25). For upper bounding $\mathbb{E}\big[e_k^2(t)\big]$, we use a propagation of chaos bound similar to (27) as follows:

$$\mathbb{E}\big[e_k(t)^2\big] \leq \left(\frac{\tilde{c}_{\mu,\nu,\tilde{G}}}{k} + \frac{\bar{c}_{\mu,\nu,\tilde{G}}}{n}\right) e^{2t\, c_{\mu,\nu,\tilde{G}}}, \tag{52}$$

where $c_{\mu,\nu,\tilde{G}}$, $\tilde{c}_{\mu,\nu,\tilde{G}}$, and $\bar{c}_{\mu,\nu,\tilde{G}}$ are as defined in (44). The proof of is similar to that of Lemma 4 and is given in Appendix A below for completeness.

From (47), (51) and (52), we have using $\sqrt{x+y} \leq \sqrt{x} + \sqrt{y}$ for all $x, y \geq 0$,

$$\mathbb{E}\left[\left|\mathbb{E}_{\nu}[e^{g_{\rho_t}}] - \mathbb{E}_{\hat{\nu}_n}[e^{g_{\bar{\rho}_{k,n,t}}}]\right|\right]$$
$$\leq \left(\frac{C_{\mathsf{D}} E\big(e^{\tilde{\mathcal{G}}}, e^{\tilde{G}}\big)}{\sqrt{n}} + \frac{\sqrt{C_{\mathsf{M}} M_2(\nu)}}{\sqrt{k}} + C_{\mathsf{L}}\sqrt{M_2(\nu)}\left(\frac{\sqrt{\bar{c}_{\mu,\nu,\tilde{G}}}}{\sqrt{n}} + \frac{\sqrt{\tilde{c}_{\mu,\nu,\tilde{G}}}}{\sqrt{k}}\right) e^{t\, c_{\mu,\nu,\tilde{G}}}\right)\left\|e^{\tilde{G}}\right\|_{2,\nu}.$$

Similarly

$$\mathbb{E}\left[\left|\mathbb{E}_{\mu}[g_{\rho_t}] - \mathbb{E}_{\hat{\mu}_n}[g_{\tilde{\rho}_{k,n,t}}]\right|\right]$$
$$\leq \left(\frac{C_{\mathsf{D}} E\big(\tilde{\mathcal{G}}, \tilde{G}\big)}{\sqrt{n}} + \frac{\sqrt{C_{\mathsf{M}} M_2(\mu)}}{\sqrt{k}} + C_{\mathsf{L}}\sqrt{M_2(\mu)}\left(\frac{\sqrt{\bar{c}_{\mu,\nu,\tilde{G}}}}{\sqrt{n}} + \frac{\sqrt{\tilde{c}_{\mu,\nu,\tilde{G}}}}{\sqrt{k}}\right) e^{t\, c_{\mu,\nu,\tilde{G}}}\right)\left\|\tilde{G}\right\|_{2,\mu},$$

from which the claim in (43) follows from (45). □

APPENDIX

A. *Proof of Equation* (52)

Set

$$b_j(\theta,\rho) \coloneqq \mathbb{E}_{\mu}\big[\partial_j\phi\big(\cdot,\theta\big)\big] - \mathbb{E}_{\nu}\big[e^{g_\rho}\partial_j\phi\big(\cdot,\theta\big)\big],$$
$$\hat{b}_{n,j}(\theta,\rho) \coloneqq \mathbb{E}_{\hat{\mu}_n}\left[\partial_j\phi\big(\cdot,\theta\big)\right] - \mathbb{E}_{\hat{\nu}_n}\left[e^{g_\rho}\partial_j\phi\big(\cdot,\theta\big)\right].$$

From (41) and (48), we have similar to (28) that

$$\frac{d}{dt}\,\mathbb{E}\big[e_k(t)^2\big] \le (1+2\beta_V)\mathbb{E}\big[e_k(t)^2\big] + \frac{1}{k}\sum_{i=1}^{k}\sum_{j=1}^{d}\mathbb{E}\Big[\big(b_j(\bar{\theta}_i(t),\rho_t) - \hat{b}_{n,j}\big(\theta_i^{(n)}(t),\tilde{\rho}_{k,n,t}\big)^2\Big]. \tag{53}$$

By Jensen's inequality,

$$\begin{aligned}
&\Big|b_j(\bar{\theta}_i(t),\rho_t) - \hat{b}_{n,j}\big(\theta_i^{(n)}(t),\tilde{\rho}_{k,n,t}\big)\Big|^2 \\
&\le 3\left|b_j(\bar{\theta}_i(t),\rho_t) - b_j\big(\bar{\theta}_i(t),\bar{\rho}_{k,t}\big)\right|^2 + 3\left|b_j\big(\bar{\theta}_i(t),\bar{\rho}_{k,t}\big) - b_j\big(\theta_i^{(n)}(t),\tilde{\rho}_{k,n,t}\big)\right|^2 \\
&\quad + 3\left|b_j\big(\theta_i^{(n)}(t),\tilde{\rho}_{k,n,t}\big) - \hat{b}_{n,j}\big(\theta_i^{(n)}(t),\tilde{\rho}_{k,n,t}\big)\right|^2.
\end{aligned} \tag{54}$$

Similar to (30),

$$\sum_{j=1}^{d}\mathbb{E}\big[\,\big|b_j(\bar{\theta}_i(t),\rho_t) - b_j\big(\bar{\theta}_i(t),\bar{\rho}_{k,t}\big)\big|^2\,\big] \le \frac{8\,C_{\mathsf{L}}^4 C_{\mathsf{M}}}{k}\big\|e^{\tilde{G}}\big\|_{2,\nu}^2 M_4(\nu). \tag{55}$$

For the last term in (54),

$$\begin{aligned}
&\Big|b_j\big(\theta_i^{(n)}(t),\tilde{\rho}_{k,n,t}\big) - \hat{b}_{n,j}\big(\theta_i^{(n)}(t),\tilde{\rho}_{k,n,t}\big)\Big|^2 \\
&\le 2\left|\mathbb{E}_{\hat{\mu}_n}\Big[\partial_j\phi\big(\cdot,\theta_i^{(n)}(t)\big)\Big] - \mathbb{E}_{\mu}\Big[\partial_j\phi\big(\cdot,\theta_i^{(n)}(t)\big)\Big]\right|^2 \\
&\quad + 2\left|\mathbb{E}_{\hat{\nu}_n}\Big[e^{g_{\tilde{\rho}_{k,n,t}}}\partial_j\phi\big(\cdot,\theta_i^{(n)}(t)\big)\Big] - \mathbb{E}_{\nu}\Big[e^{g_{\tilde{\rho}_{k,n,t}}}\partial_j\phi\big(\cdot,\theta_i^{(n)}(t)\big)\Big]\right|^2.
\end{aligned}$$

Then, similar to (32),

$$\sum_{j=1}^{d}\mathbb{E}\left[\left|\mathbb{E}_{\hat{\mu}_n}\Big[\partial_j\phi\big(\cdot,\theta_i^{(n)}(t)\big)\Big] - \mathbb{E}_{\mu}\Big[\partial_j\phi\big(\cdot,\theta_i^{(n)}(t)\big)\Big]\right|^2\right] \le \frac{C_{\mathsf{L}}^2 M_2(\mu)}{n},$$

$$\sum_{j=1}^{d}\mathbb{E}\left[\left|\mathbb{E}_{\hat{\nu}_n}\Big[e^{g_{\tilde{\rho}_{k,n,t}}}\partial_j\phi\big(\cdot,\theta_i^{(n)}(t)\big)\Big] - \mathbb{E}_{\nu}\Big[e^{g_{\tilde{\rho}_{k,n,t}}}\partial_j\phi\big(\cdot,\theta_i^{(n)}(t)\big)\Big]\right|^2\right] \le \frac{C_{\mathsf{L}}^2}{n}\mathbb{E}_{\nu}\left[e^{2\tilde{G}(Y)}\,\|Y\|_2^2\right].$$

From the above,

$$\sum_{j=1}^{d}\mathbb{E}\left[\left|b_j\big(\theta_i^{(n)}(t),\tilde{\rho}_{k,n,t}\big) - \hat{b}_{n,j}\big(\theta_i^{(n)}(t),\tilde{\rho}_{k,n,t}\big)\right|^2\right] \le \frac{2C_{\mathsf{L}}^2 M_2(\mu)}{n} + 2\frac{C_{\mathsf{L}}^2}{n}\mathbb{E}_{\nu}\left[e^{2\tilde{G}(Y)}\,\|Y\|_2^2\right]. \tag{56}$$

Finally, for the second term in (54), we obtain similar to (33) using $(x+y+z)^2 \leq 3x^2+3y^2+3z^2$ for $x, y, z \in \mathbb{R}$:

$$\sum_{j=1}^{d} \mathbb{E}\left[\left|b_j\big(\bar{\theta}_i(t), \bar{\rho}_{k,t}\big) - b_j\big(\theta_i^{(n)}(t), \tilde{\rho}_{k,n,t}\big)\right|^2\right]$$
$$\leq 3C_{\mathsf{L}}^2\Big(M_2(\mu) + \big\|e^{\tilde{G}}\big\|_{2,\nu}^2 M_2(\nu)\Big)\mathbb{E}\left[\big\|\bar{\theta}_i(t) - \theta_i^{(n)}(t)\big\|_2^2\right] + 3C_{\mathsf{L}}^4\big\|e^{2\tilde{G}}\big\|_{2,\nu}\sqrt{M_4(\nu)}M_2(\nu)\mathbb{E}\big[e_k^2\big].$$

From (53), (54), (55), (56) and the above equation, we have

$$\frac{d}{dt}\mathbb{E}\big[e_k(t)^2\big] \leq \Big(1 + 2\beta_V + 9C_{\mathsf{L}}^2\Big(M_2(\mu) + \big\|e^{\tilde{G}}\big\|_{2,\nu}^2 M_2(\nu)\Big) + 9C_{\mathsf{L}}^4\big\|e^{2\tilde{G}}\big\|_{2,\nu}\sqrt{M_4(\nu)}M_2(\nu)\Big)\,\mathbb{E}\big[e_k(t)^2\big]$$
$$+ \frac{24\,C_{\mathsf{L}}^4 C_{\mathsf{M}}}{k}\big\|e^{\tilde{G}}\big\|_{2,\nu}^2 M_4(\nu) + \frac{6C_{\mathsf{L}}^2 M_2(\mu)}{n} + \frac{6C_{\mathsf{L}}^2}{n}\big\|e^{2\tilde{G}}\big\|_{2,\nu}\sqrt{M_4(\nu)}.$$

The claim in (52) then follows via an application of Gronwall's lemma (Lemma 5).

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